

Novel Multi Source Inverter for Electric Vehicles - Final Demo

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Introduction

- Growing demand for efficient EV power electronics in the last decade
- Limitations of current inverters:
 1. Single DC source => limited flexibility
 2. High switching losses & harmonics
 3. Large size / high component count
- Trend toward multi-source energy systems in EVs, needed for smarter, more adaptable inverter designs
- Our solution: Novel Multi-Source Inverter (MSI)
- Combines multiple DC inputs
- Improves efficiency & performance
- Reduces complexity and size



Problem Statement, Existing Solutions, Project Objectives

Problem Statement

- EV inverters don't operate at peak efficiency across all load conditions
- Efficiency drops significantly at partial loads (which is most real-world driving)
- This directly impacts vehicle range and energy utilization
- Most EVs rely on a single battery pack as the DC source
- Limited flexibility in how power is delivered to the motor
- Cannot adapt voltage/current dynamically to optimize efficiency
- Forces the inverter to handle a wide operating range inefficiently

Existing Solutions

Current Solutions in the Market:

- No current market solution exists, however this is a very active area of EV research.
- Reconfigurable batteries exist, but is used as a single battery system that dynamically switches from series and parallel cells.
- Existing designs require many switches, leading to more complexity, and increased cost.
- Our project aims to develop a solution that maintains the benefit of multi-source inverters, while reducing component count and complexity.

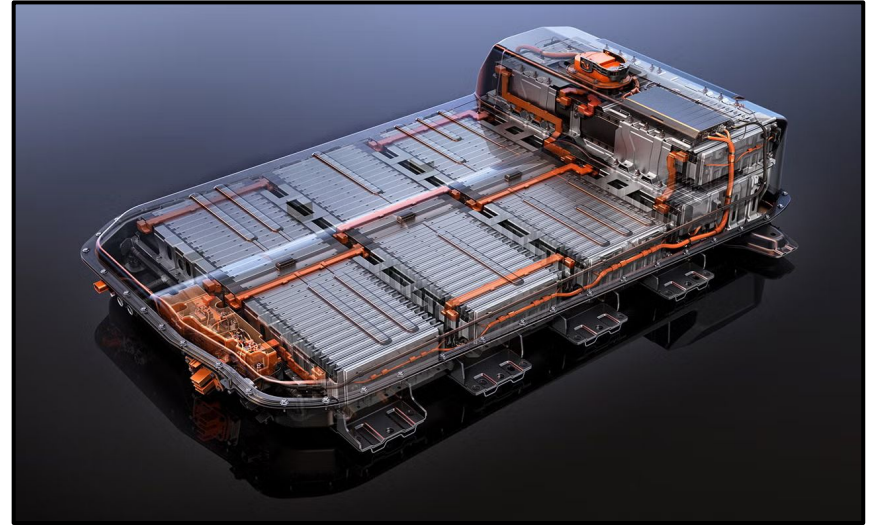


Figure 1. Modern EV Battery System

Project Objectives

Our project objectives are closely tied to our requirements.

- Minimize component count
- Develop a PCB with a minimal footprint
- Design a novel multi-source input
- Meet safety requirements
- Meet budget requirement of \$800
- Minimize harmonic distortion

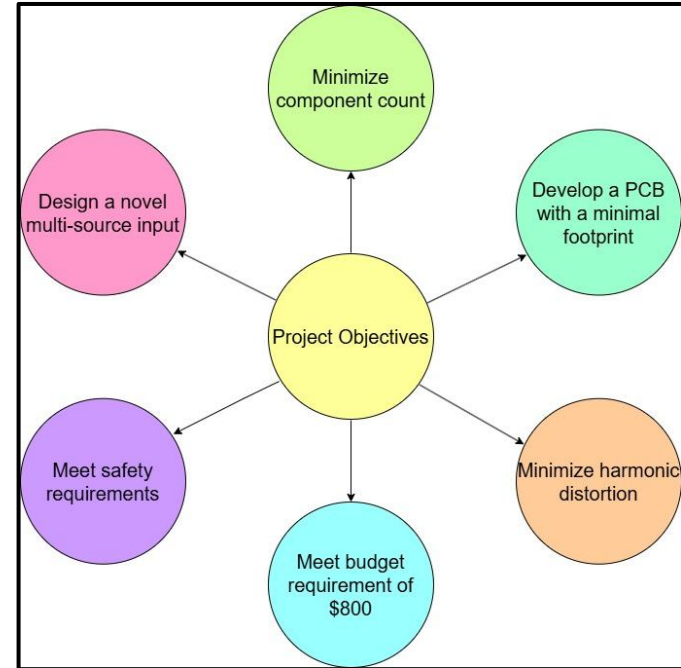


Figure 2. Project Objectives

Engineering Design

Design Process

- **Iterative engineering workflow**
- Problem Definition
- Requirement Specification
- Research
- Concept generation
- Simulations
- Conceptual system Design
- PCB Layout
- Firmware Design
- Testing and Evaluation

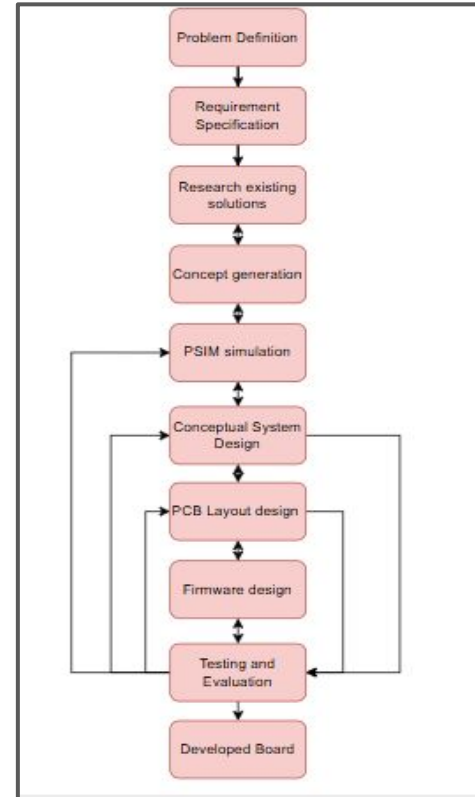


Figure 3. Design Process

Proposed Solution - Novel Design

MSI Novel Design:

- Two battery inputs
- Parallel design
- Minimized component count
- Multiple voltage levels
- Can be scaled to full size system
- Application adjustments can be made

Challenges and Concerns:

- In a full scale model, diodes can be replaced with IGBT/SiC MOSFETs for active control
- Shoot-through can cause extensive damage
- Current needs to be controlled when switching modes

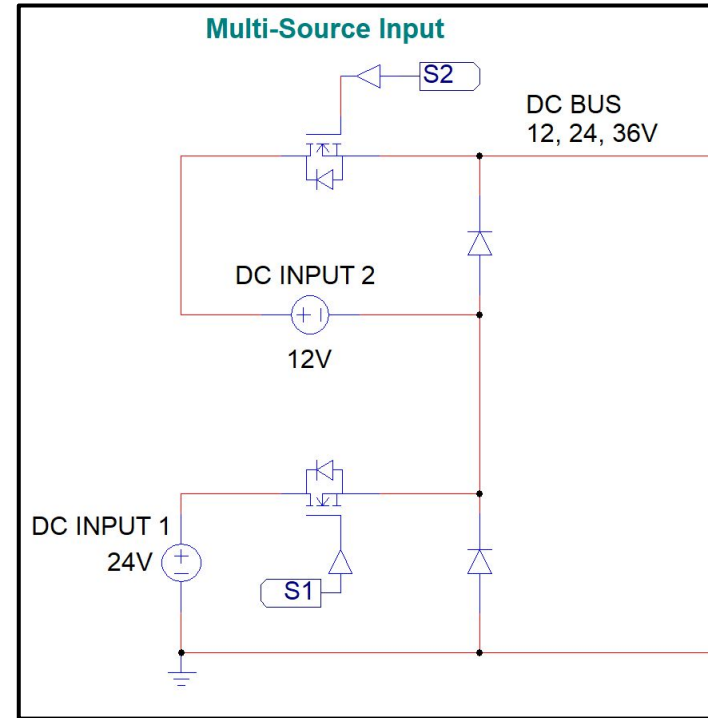


Figure 4. Novel MSI Design

Proposed Solution - Novel Design

Proposed Solution: Multi-Source Inverter (MSI)

- Integrates multiple DC inputs into one system

Key architecture blocks:

- Multi-source input switching
- DC-DC regulation stage
- STM32 MCU (control + PWM)
- 3-phase inverter topology
- Feedback sensors (closed-loop control)

Key design goals:

- Dynamic source selection
- Improved efficiency (low & high speed)
- Reduced component count
- Modular & scalable architecture

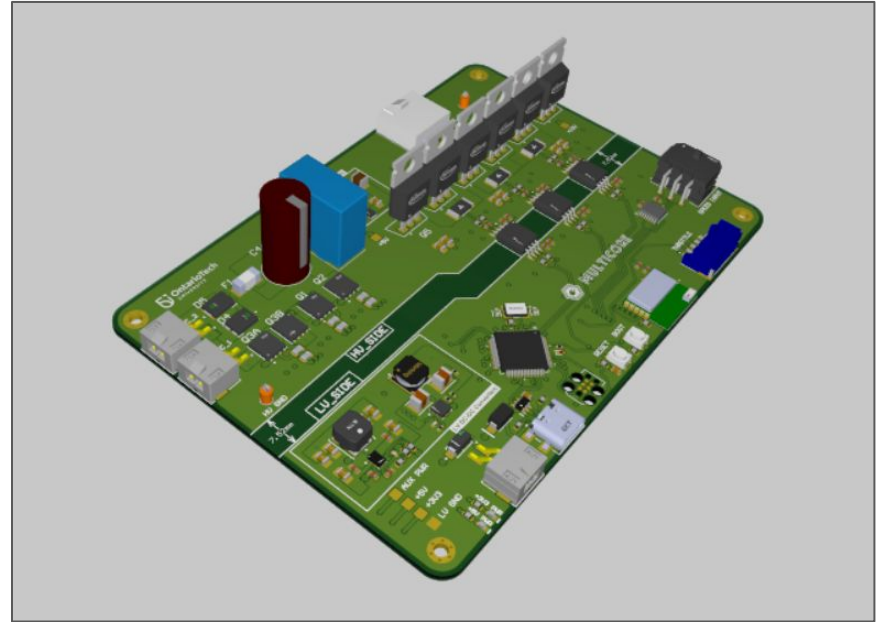
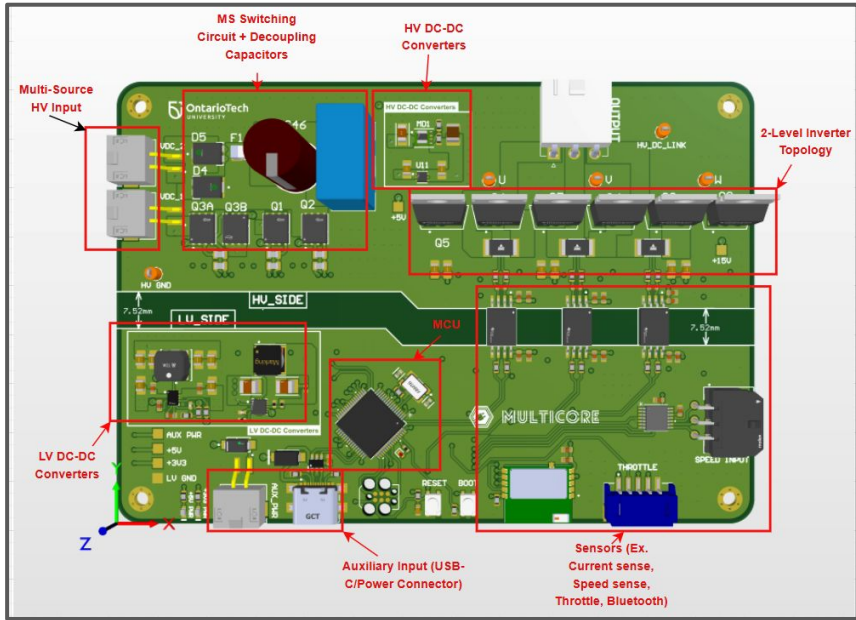


Figure 5. Final Design Solution

Final Product

Top Side



Bottom Side

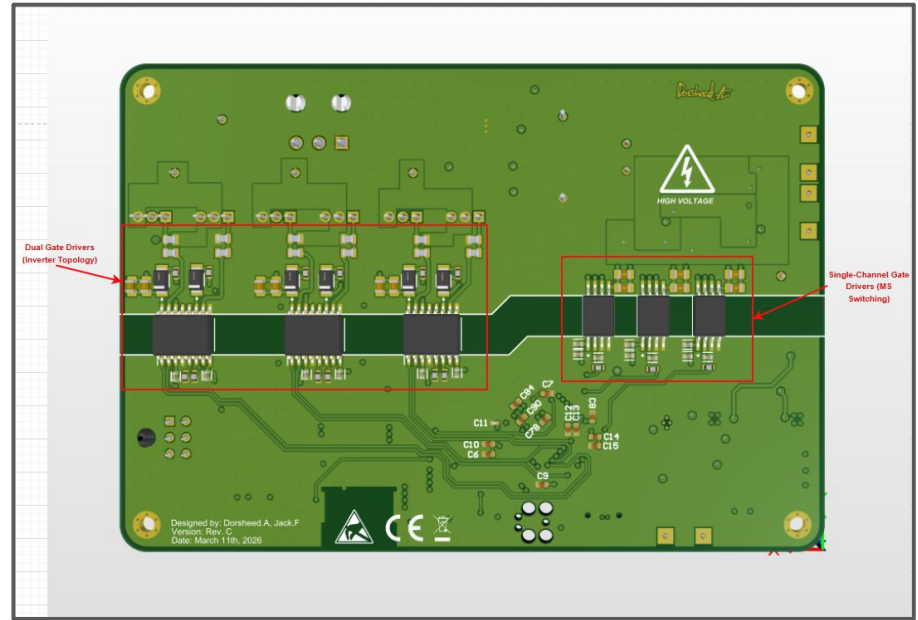
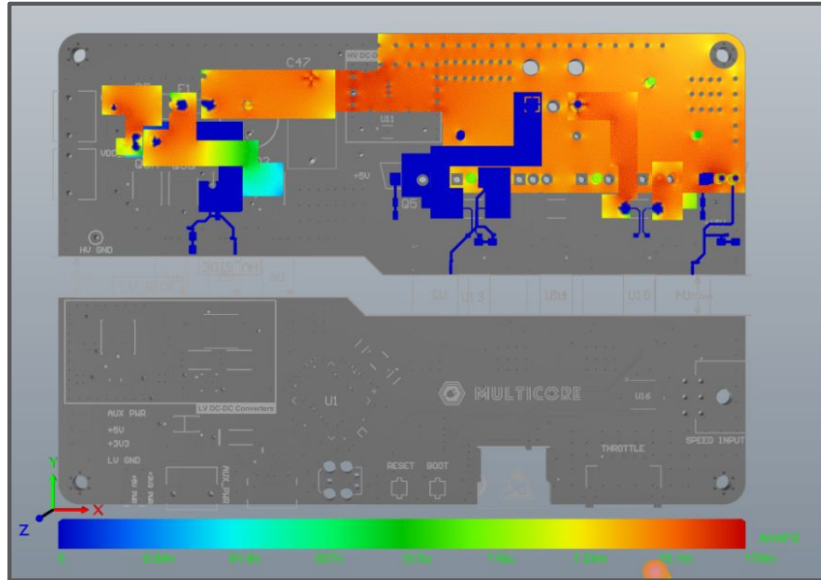


Figure 6. Final PCB Layout

Final Product

Power Delivery Simulation



Assembled Board

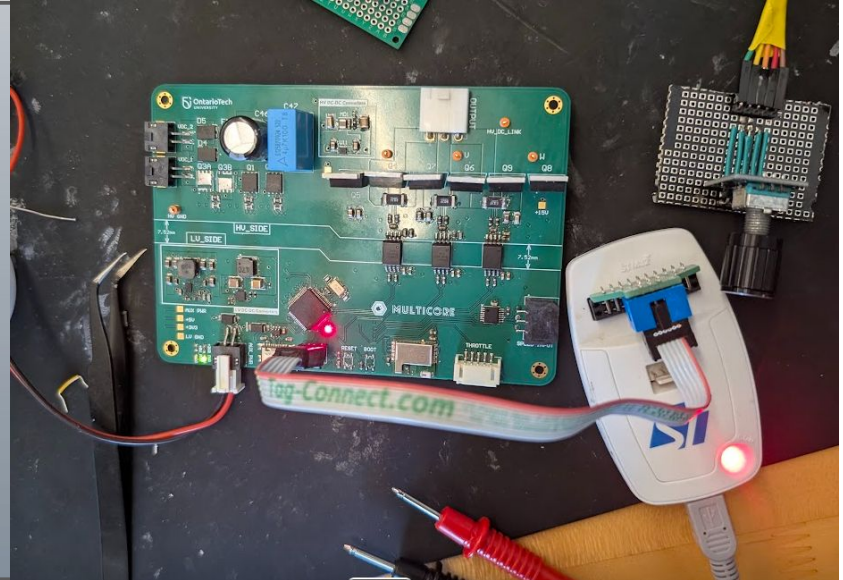


Figure 7. Final Product

Final Product

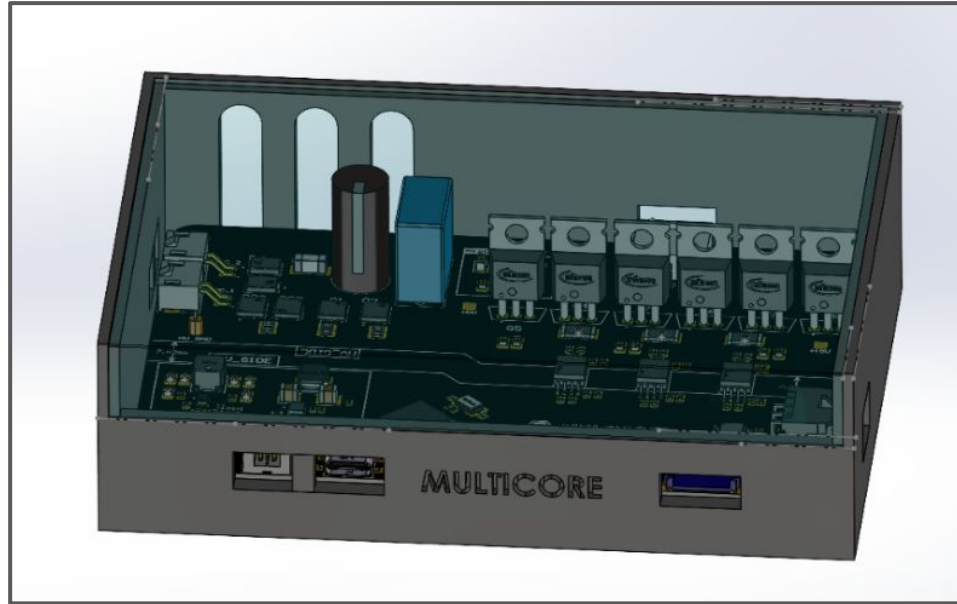


Figure 8. Final Product

MSI Board with Enclosure

Estimated Project Budget

The table below highlights estimated budget of key materials for our project

Materials	Cost per unit(\$)	Quantity	Total Cost (\$)
Motor	109.33	1	109.33
MOSFETS	10.00	6	60.00
PCB Fabrication	80.00	1	80.00
18650 li-on battery	17.00	1	17.00
STM32G491VET6 board	35.26	1	35.26
Most PCB components			159.46
Total			461.05

Actual Project Budget

Materials	Cost per unit(\$)	Quantity	Total Cost (\$)
Motor	73.56	1	73.56
PCB Fabrication	66.49	3	199.47
Batteries(12V and 24V	94.13	1	94.13
Most PCB components			489.61
Total			856.77

Analysis & Testing

Evaluation Results and Analysis - Methodology

Testing Approach: From individual component validation to integration testing.

Tools and Metrics:

- Hardware: Multimeter probing for power rails; Oscilloscope for PWM.
- Software: STM32IDE Live expressions to verify current sensors and throttle.
- Simulation: Altium PDN analyzer for current density and voltage drop.

Subsystems Tested: Power management, Gate drivers, Regulators, Throttle control, and MCU logic.

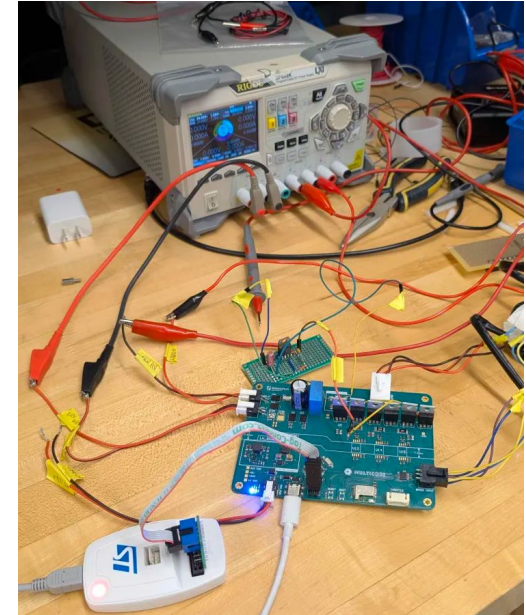


Figure 9. Testing Setup

Quantitative Results & Requirement comparison

- PWM Performance
- PDN Analysis
- Physical Design

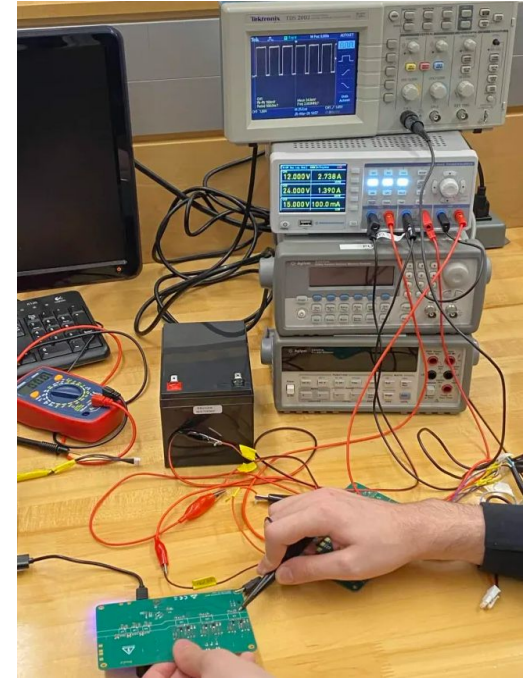
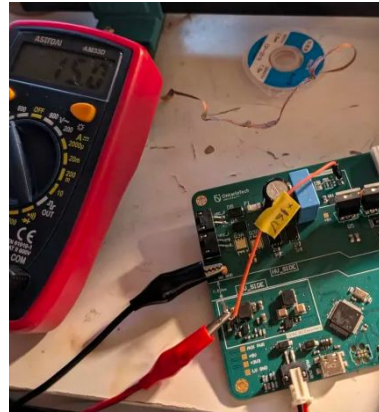
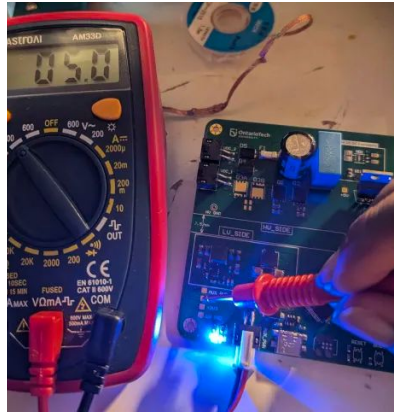


Figure 10. PCB Testing

Quantitative Results & Requirement comparison pt 2

Target vs Result Check:

- Component Efficiency: Met.
- Physical footprint: Met.
- Power and Frequency: Met.
- Source switching: Failed.
- Budget: Failed. (due to component and design failure)
- Performance Metrics: Failed.
- Simulation Validation: Met.



Challenges & Solutions

Challenges:

1. PCB Design Defect
2. Communication Blockers
3. DC-DC Converter
4. Firmware implementation
5. Shorts in the circuit
6. 5A Fuse

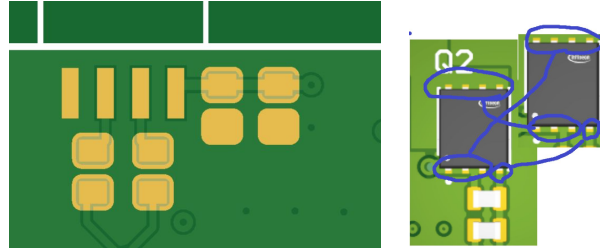


Figure 11. PCB Defect



Figure 12. DCDC Converter

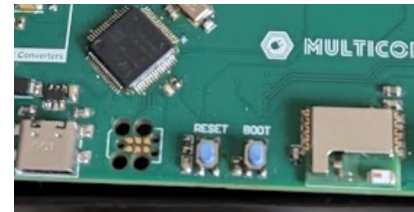


Figure 13. USB-C and Bluetooth



Figure 14. Fuse

Challenges & Solutions pt 2

Solutions:

1. Fixing design
2. Using STM32 STLINK
3. Voltage Regulator replacement
4. Cursor aided in fixing errors
5. Fixed excess solder
6. 10A Fuse

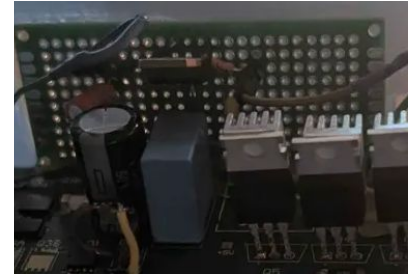
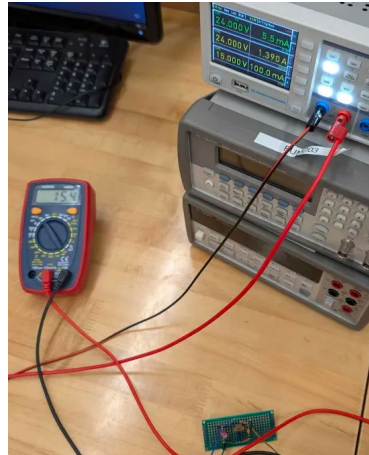
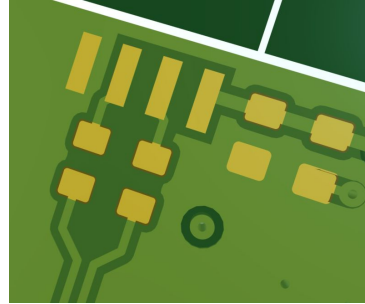


Figure 14. Solutions

Root Cause Analysis & Limitations

- **Switching Failure:** During MSI mode switching (changing DC voltage levels), a MOSFET burned out.
- **Testing Constraints:** The prototype was limited to low-voltage (max 36V) and did not achieve full SVM modulation on hardware due to PCB defects.

Assumptions: Testing components would meet datasheet specs without parasitic interference, which was later proven false by the PCB partial integration test.

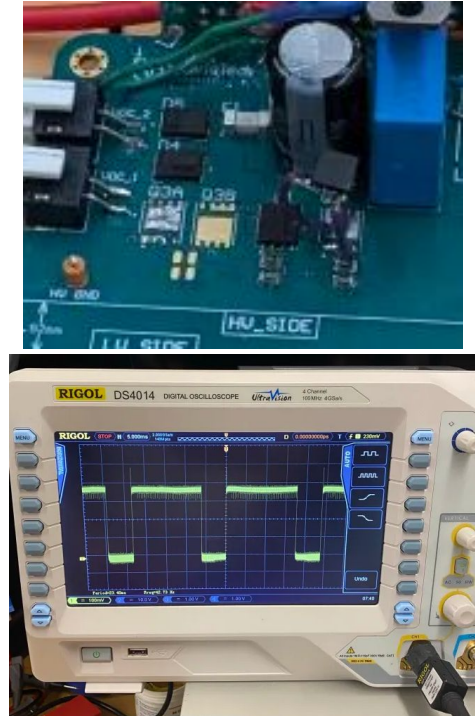


Figure 15. Causes and limitations

Conclusion & Future Work



Q&A

